

and unpack it with `tar xvfz qucs-.tar.gz`; then switch to the source directory (`cd qucs-<version>`) and configure the source package for your system with `./configure`. Compile the package with `make` and install it with `make install`.

Once installed, run `qucs` in the terminal or launch it from the Unity dashboard.

The QUCS structure

QUCS is actually a collection of different modules, each with a different function:

- The GUI is used to create schematics, set up simulations, display simulation results, for writing VHDL code, etc.
- The analogue simulator is a command line program, which is run by the GUI in order to simulate the schematic you previously set up. It takes a netlist, checks it for errors, performs the required simulation actions, and finally produces a dataset.
- The text editor is used to display netlists and simulation logging information, and to edit files included by certain components (like SPICE netlists or Touchstone files).
- The filter synthesis application can be used to design various types of filters.
- The transmission line calculator can be used to design and analyse different types of transmission lines (like microstrips and coaxial cables).
- The component library manager holds models for real-life devices (like transistors, diodes, bridges and opamps). It can be extended by the user.
- The attenuator synthesis application can be used to design various types of passive attenuators.
- The command line conversion program tool is used by the GUI to import and export datasets, netlists and schematics from and to other CAD/EDA software. The supported file formats as well as usage information can be found in the `qucsconv` man page.
- Additionally, the GUI steers other EDA tools. For digital simulations (via VHDL), the program FreeHDL is used. For circuit optimisation (minimisation of a cost function), ASCO is configured and run.

The QUCS component library has almost all the components needed for most of the circuits, but as it is still in the development stages, some components might be missing. The following categories of components are provided:

- Lumped components (R, L, C, amplifier, phase shifter, etc)
- Sources
- Probes
- Transmission lines
- Non-linear components (diodes, transistors, etc)
- Digital components
- File containers (S-parameter datasets and SPICE netlists)
- Paintings

There is also a component library that includes various

standard components available in the market (bridges, diodes, varistors, LEDs, JFETs, MOSFETs, and so on).

Working with QUCS

So let's start off with some circuits. Launch QUCS and proceed to start a new project. Click the *New* button in the explorer window; a dialogue box will appear asking for the project name—supply one, and proceed. Once the project is created, you can see all the contents of the project in the Explorer window. The workspace is where you draw your circuit.

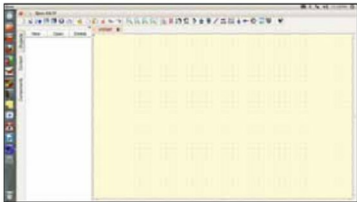


Figure 1: The QUCS main window

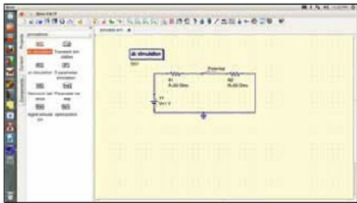


Figure 2: A circuit ready for simulation

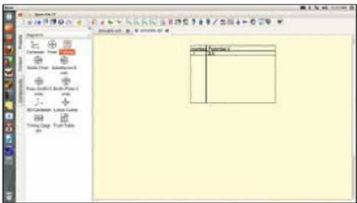


Figure 3: A tabular output

Let's start with a simple potential divider circuit. You can find the components via the components button on the